PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

WEEK 02 - NETWORKWALKS INTERNSHIP PROGRAMME

Particulars	Details
Pentester Name	Subhangi
Program / Batch	B084 – Networkwalks
Modules Completed	W2-PM1 – Multiple Kali Tools
	W2-PM5 – Zenmap Scanning
Client / Target	1. Networkwalks (secured written permission already) 2. Own LAN Network
Permission Secured from Client	Yes
Phases Covered	Phase 1: Reconnaissance & Footprinting
	Phase 2: Scanning & Network Discovery
Current Progress	Phases 3–5: In Progress

TABLE OF CONTENTS

1. Liability Disclaimer
2. Introduction
3. Objectives
4. Tools and Technologies Used
5. W2-PM1: Footprinting & Reconnaissance Using Kali Linux
    - 5.1 WHOIS Enumeration
    - 5.2 Web Technology Fingerprinting Using WhatWeb
    - 5.3 DNS Resolution Using Nslookup
    - 5.4 HTTP Header Analysis Using Curl
    - 5.5 Web Application Firewall Detection Using Wafw00f
    - 5.6 DNS Enumeration Using DNSRecon
6. W2-PM5: Network Scanning Using Zenmap
    - 6.1 Network Configuration
    - 6.2 Methodology

- 6.3 Live Host Discovery
- 6.4 MAC Address Identification
- 6.5 Network Topology

7. Risk Analysis
8. Security Recommendations
9. Conclusion
10. Evidences and Screenshots

1. LIABILITY DISCLAIMER

This report was written as part of the Networkwalks Internship Program's second week of practical exercises.

The activities included in this report were carried out for cybersecurity training and instructional reasons. While network scanning was done on the local network, the footprinting and reconnaissance operations concentrated on publicly accessible information related to the internship domain.

The results are based on observations made during the practical exercises and should not be used as proof of security flaws without more investigation.

This report's contents are meant for educational and internship assessment purposes.

2. INTRODUCTION

Footprinting, reconnaissance, and network scanning are crucial stages in cybersecurity evaluations. They enable security experts to gain insights into the infrastructure, technologies, and network settings connected to a target.

During Week 2 of the Networkwalks Internship Programme, hands-on exercises were conducted using Kali Linux and Zenmap to learn about information gathering, web technology recognition, DNS enumeration, and network discovery.

The first hands-on module, W2-PM1, concentrated on gathering publicly accessible information regarding networkwalks.com, utilizing various reconnaissance tools available in Kali Linux.

The second hands-on module, W2-PM5, focused on network scanning and host discovery through Zenmap on a local Wi-Fi network. This exercise entailed detecting active hosts, noting IP and MAC addresses, and creating a network topology. These tasks offered practical experience with essential penetration testing methods and network reconnaissance strategies.

3. OBJECTIVES

The major objectives of the practical exercises were:

1. To understand the fundamentals of footprinting and reconnaissance.
2. To collect publicly available domain registration information.
3. To identify web technologies and application components.
4. To perform DNS resolution and enumerate DNS records.
5. To examine HTTP response headers.
6. To identify detectable Web Application Firewall technologies.

7. To understand network scanning using Zenmap.
8. To identify active devices within a local network.
9. To record IP addresses and available MAC address information.
10. To visualize network topology and document the findings.

4. TOOLS AND TECHNOLOGIES USED

Tool / Technology	Purpose
Kali Linux	Penetration testing environment
WHOIS	Domain registration information
WhatWeb	Web technology fingerprinting
Nslookup	DNS resolution
Curl	HTTP header analysis
Wafw00f	Web Application Firewall detection
DNSRecon	DNS enumeration
Zenmap	Graphical network scanning interface
Nmap	Network discovery and scanning engine
Windows 11	Local network configuration
Command Prompt	Network configuration identification

5. W2-PM1: FOOTPRINTING & RECONNAISSANCE USING KALI LINUX

5.1 WHOIS Enumeration

Tool Used: WHOIS

Target: networkwalks.com

Objective:

To obtain publicly available domain registration information, including registrar details, domain creation and expiry dates, domain status and name servers.

Command Executed:

```
whois networkwalks.com
```

Observations:

The WHOIS query returned the following information:

Parameter	**Result**
Domain Name	networkwalks.com
Registrar	GoDaddy.com, LLC
Domain Creation Date	6 November 2019
Updated Date	12 November 2025
Registry Expiry Date	6 November 2027
Registrar IANA ID	146
DNSSEC	Unsigned

Registrant Organization	Domains By Proxy, LLC
Registrant Privacy	Enabled

Name Servers Identified:

1. NS6135.HOSTGATOR.COM
2. NS6136.HOSTGATOR.COM
3. NS29.DOMAINCONTROL.COM
4. NS30.DOMAINCONTROL.COM

Domain Status:

The following domain status restrictions were identified:

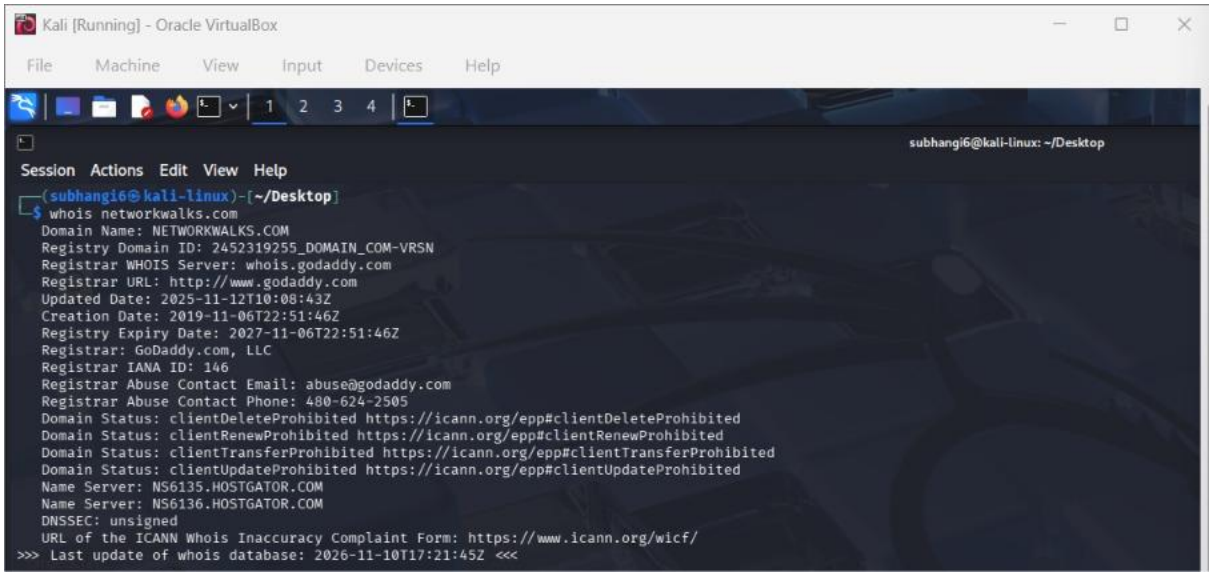
- clientDeleteProhibited
- clientRenewProhibited
- clientTransferProhibited
- clientUpdateProhibited

Security Significance:

WHOIS enumeration enables security experts to gather publicly accessible information related to domain registrations and to gain insights into fundamental domain infrastructure.

It can aid in revealing details about domain ownership structures, registrar details, and name server settings during the reconnaissance stage.

Figure 1: WHOIS enumeration using Kali Linux.



5.2 Web Technology Fingerprinting Using WhatWeb

Tool Used: WhatWeb

Target: networkwalks.com

Objective:

To identify publicly detectable technologies, frameworks, content management systems and other components used by the target website.

Command Executed:

```
whatweb networkwalks.com
```

Observations:

The WhatWeb scan initially returned an HTTP 301 redirection and subsequently identified the HTTPS website with an HTTP 200 OK response.

The following technologies and components were detected:

Parameter	Detected Result
Web Server	Apache
Website IP	192.232.216.135
Content Management System	WordPress
WordPress Version	7.1.4
WordPress Download Manager	3.3.76
Bootstrap	7.1.4
jQuery	3.7.1
Google Tag Manager	Detected
HTML5	Detected
Website Title	Networkwalks Academy
Public Email	info@networkwalks.com
HTTP Status	200 OK

The scan also identified the cookie named __wpdm_client and several uncommon HTTP response headers.

Security Significance:

Web technology fingerprinting aids security experts in recognizing the software elements and frameworks utilized by a website. This knowledge can be beneficial for examining software versions, pinpointing obsolete components, and assessing the overall technology stack during an authorized security review. Just identifying a technology does not confirm that a component is susceptible to vulnerabilities.

Figure 2: Web technology fingerprinting using WhatWeb.



5.3 DNS Resolution Using Nslookup

Tool Used: Nslookup

Target: networkwalks.com

Objective:

To resolve the target domain name into its corresponding IPv4 address and identify the DNS resolver used.

Command Executed:

```
nslookup networkwalks.com
```

Observations:

The command returned the following information:

Parameter	**Result**
DNS Server	192.168.1.1
DNS Port	53
Response Type	Non-authoritative answer
Domain Name	networkwalks.com
Resolved IPv4 Address	192.232.216.135

The DNS query was processed through the local network DNS resolver at 192.168.1.1.

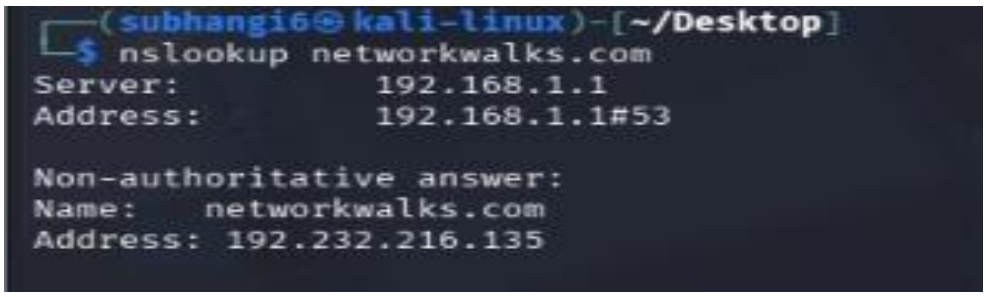
The domain networkwalks.com successfully resolved to IPv4 address 192.232.216.135.

Security Significance:

DNS resolution is an important part of reconnaissance because it helps security professionals identify the IP address associated with a domain.

This information supports infrastructure identification and further authorized security assessment activities.

Figure 3: DNS resolution using Nslookup.



5.4 HTTP Header Analysis Using Curl

Tool Used: Curl

Target: https://networkwalks.com

Objective:

To inspect HTTP response headers and identify publicly exposed web server information, application metadata, cookie attributes and security-related headers.

Command Executed:

```
curl -I https://networkwalks.com
```

Observations:

The website returned an HTTP/2 200 response, indicating that the request was successful.

The following HTTP headers were identified:

Header	**Observed Value**
HTTP Status	HTTP/2 200
Server	Apache
Content-Type	text/html; charset=UTF-8
Date	10 November 2026
Referrer-Policy	no-referrer-when-downgrade
X-Nginx-Cache	WordPress
X-Endurance-Cache-Level	0
Cookie	__wpdm_client
Cookie Attributes	Secure, HttpOnly

The response also included a WordPress REST API discovery link:

```
https://networkwalks.com/wp-json/
```

An additional API reference was identified:

`https://networkwalks.com/wp-json/wp/v2/pages/53`

The cookie was observed with Secure and HttpOnly attributes.

Security Significance:

HTTP header analysis helps identify web server information, application technologies, cookie configurations and security-related response policies.

The presence of application metadata and API references can help security professionals understand the web application's publicly exposed interface.

The existence of a REST API discovery link does not automatically indicate a security vulnerability.

Figure 4: HTTP response header analysis using Curl.



5.5 Web Application Firewall Detection Using Wafw00f

Tool Used: Wafw00f

Target: https://networkwalks.com

Objective:

To identify whether a Web Application Firewall is detectable on the target website.

Command Executed:

```
wafw00f https://networkwalks.com
```

Observations:

The Wafw00f scan successfully checked the target website.

The output identified:

Parameter	**Result**
WAF Detection	Successful
Detected WAF	ModSecurity
Provider	SpiderLabs
Number of Requests	2

The tool reported that the website was behind a ModSecurity (SpiderLabs) Web Application Firewall.

Security Significance:

A Web Application Firewall helps protect web applications by inspecting incoming HTTP traffic and applying security rules.

WAF detection helps security professionals understand the presence of protective technologies within a web application environment.

The detection of a WAF does not establish its effectiveness or guarantee protection against all attacks.

Figure 5: Web Application Firewall detection using Wafw00f.



5.6 DNS Enumeration Using DNSRecon

Tool Used: DNSRecon

Target: networkwalks.com

Objective:

To enumerate publicly available DNS records and identify information about the target domain's DNS infrastructure and associated services.

Command Executed:

```
dnsrecon -d networkwalks.com
```

Observations:

DNSRecon successfully performed general DNS enumeration and identified multiple DNS record types.

The following records were observed:

A. SOA Record

```
SOA ns6135.hostgator.com 50.87.144.87
```

The Start of Authority record identified ns6135.hostgator.com as the primary name server listed in the SOA response.

B. NS Records

The following name servers were identified:

Name Server	IP Address
ns6136.hostgator.com	192.232.216.131
ns6135.hostgator.com	50.87.144.87

C. DNS Server Version Information

DNSRecon reported the following BIND version strings during its DNS server version checks:

Server IP	Reported Version
192.232.216.131	9.16.23-RH
50.87.144.87	9.16.23-RH

D. MX Record

```
MX mail.networkwalks.com 192.232.216.135
```

The MX record identified mail.networkwalks.com as a mail server hostname, with the corresponding IP address shown in the output.

E. A Record

```
A networkwalks.com 192.232.216.135
```

The A record mapped the domain name to its IPv4 address.

F. TXT Records

Two TXT records were identified:

1. Google site verification record.
2. Sender Policy Framework (SPF) record.

The SPF record observed was:

v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all

This record specifies permitted sources for sending email on behalf of the domain.

G. SRV Records

DNSRecon identified eight SRV records associated with:

`_autodiscover._tcp.networkwalks.com`

These records pointed to cPanel email autodiscovery infrastructure.

H. DNSSEC Observation

DNSRecon returned the following message:

ERROR No answer for DNSSEC query for networkwalks.com

The WHOIS output also reported DNSSEC as unsigned.

Summary of DNS Enumeration

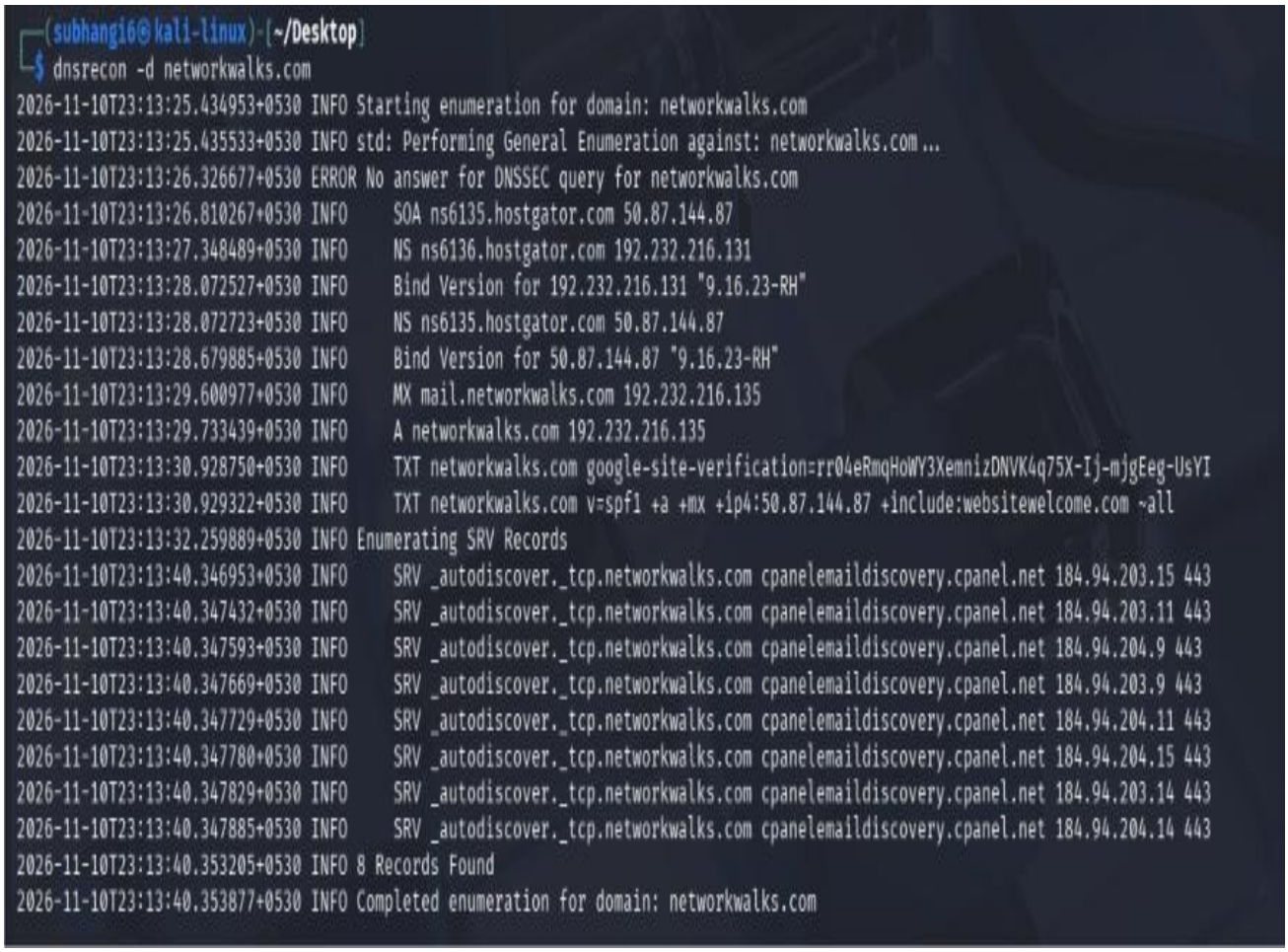
Record Type	**Result**
SOA	Identified
NS	2 records
MX	Identified
A	Identified
TXT	2 records
SRV	8 records
DNSSEC	No answer received to DNSSEC query

Security Significance:

DNS enumeration helps security professionals understand domain infrastructure, identify mail services, review DNS configurations and collect information about publicly exposed service records.

The findings can assist in evaluating DNS configuration and identifying information that should be reviewed as part of infrastructure security maintenance.

Figure 6: DNS enumeration using DNSRecon.



6. W2-PM5: NETWORK SCANNING USING ZENMAP

6.1 Objective

The objective of this practical was to identify active hosts within a local area network using Zenmap, determine their IP and MAC addresses where available, and visualize the network topology.

6.2 Tools and Technologies Used

The following tools and technologies were used:

- Nmap 7.993.
- Zenmap GUI.
- Windows 11.
- Windows Command Prompt.

6.3 Network Configuration

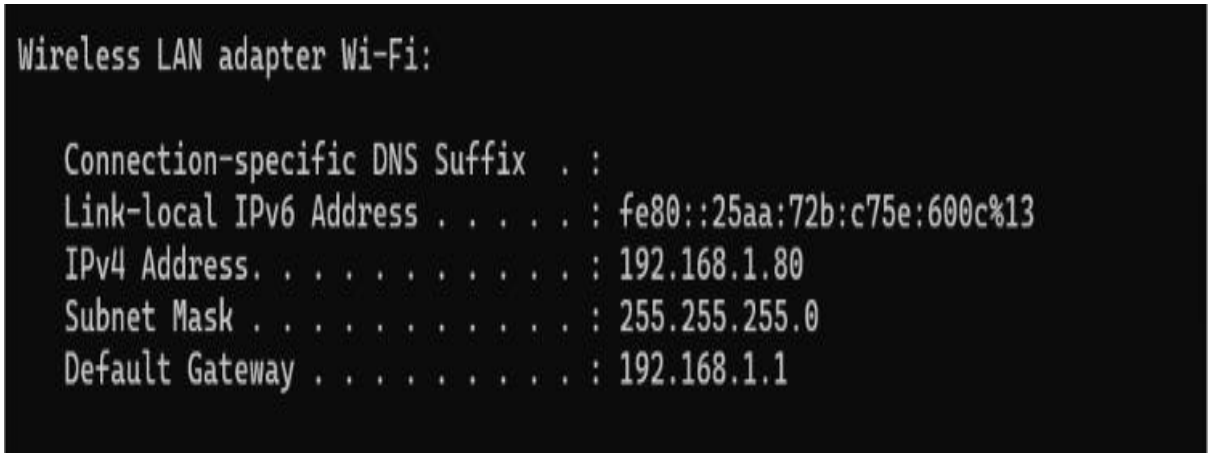
The Windows `ipconfig /all` command was used to identify the active Wi-Fi adapter and local network configuration.

The following details were recorded:

Parameter	**Value**
Network Adapter	Realtek RTL8852BE Wi-Fi 6
IPv4 Address	192.168.1.80
Subnet Mask	255.255.255.0
Network Subnet	192.168.1.0/24
Default Gateway	192.168.1.1
MAC Address	C0-BF-BE-C8-CC-1A

The local subnet was identified as 192.168.1.0/24.

Figure 7: Local network configuration using Windows Command Prompt.

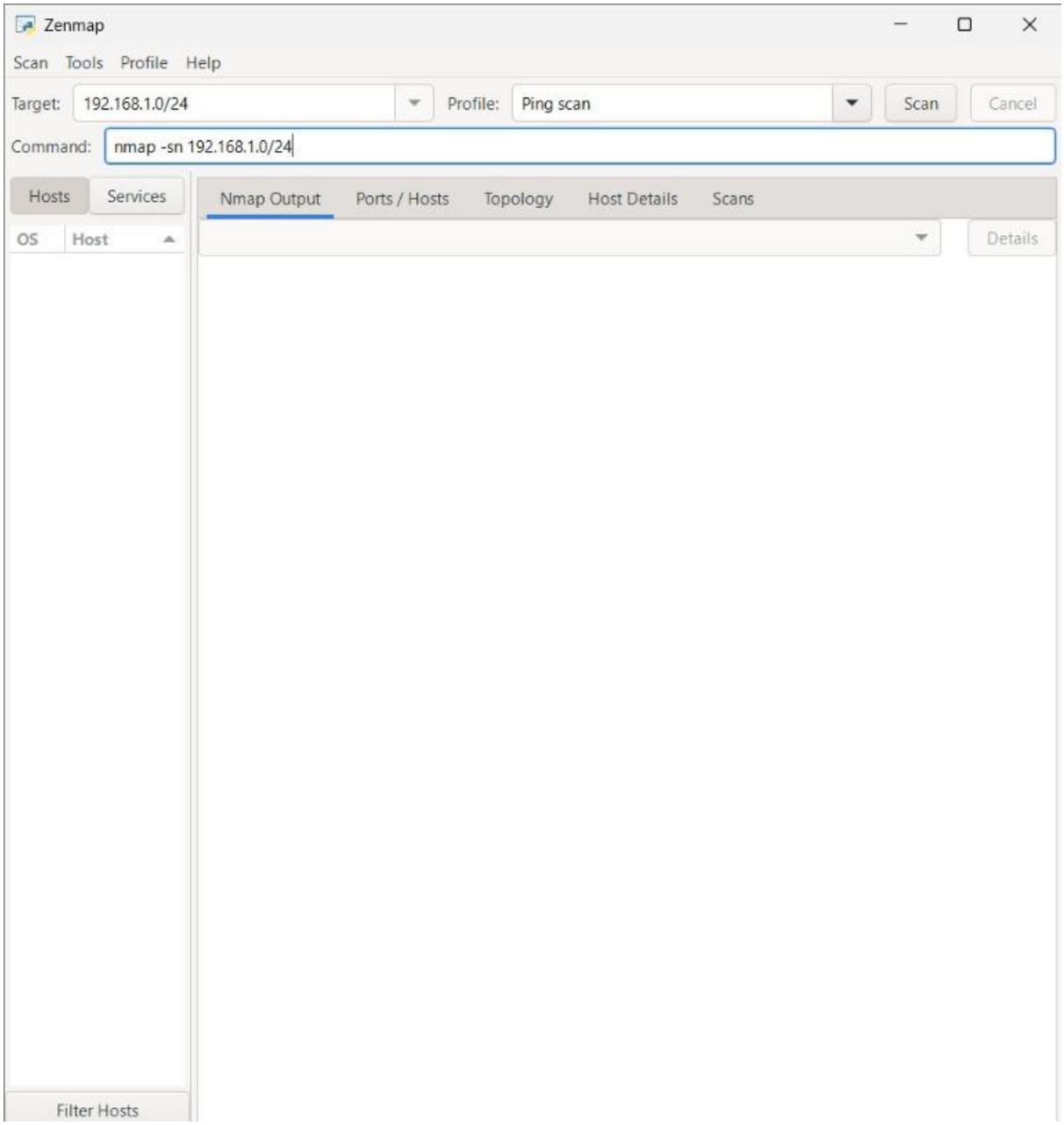


6.4 Methodology

The network scanning practical was performed using the following steps:

1. Downloaded and installed Zenmap from the official Nmap website.
2. Identified the active Wi-Fi adapter and local IP configuration using the ipconfig command.
3. Determined the local network subnet as 192.168.1.0/24.
4. Configured Zenmap with the target subnet.
5. Selected the Ping Scan profile.
6. Executed the network discovery scan.
7. Recorded the active hosts and their IP addresses.
8. Identified available MAC address information using Nmap output and Windows network configuration.
9. Generated the network topology using Zenmap.

Figure 8: Zenmap scan configuration.



6.5 Live Host Discovery Results

The Nmap scan examined 256 IP addresses within the local network and identified five active hosts.

Total Live Hosts Discovered: 5

The following results were recorded:

No.	IP Address	MAC Address
1	192.168.1.1	98:35:ED:A2
2	192.168.1.3	EC:0D:E4:63:DC:83
3	192.168.1.26	D6:D6:DF:01:0F:37
4	192.168.1.173	D6:D6:DF:01:0F:37
5	192.168.1.80	C0-BF-BE-C8-CC-1A

The first device was identified as Huawei Technologies, while the second device was identified as Amazon Technologies.

The remaining devices were reported as unknown by Nmap.

The MAC address of the local computer was obtained separately through the Windows `ipconfig /all` command.

The first MAC address should be verified against the original scan output because it was recorded with five octets rather than the conventional six. The repeated MAC address for two hosts should also be checked against the original evidence.

Figure 9: Live host discovery using Zenmap.

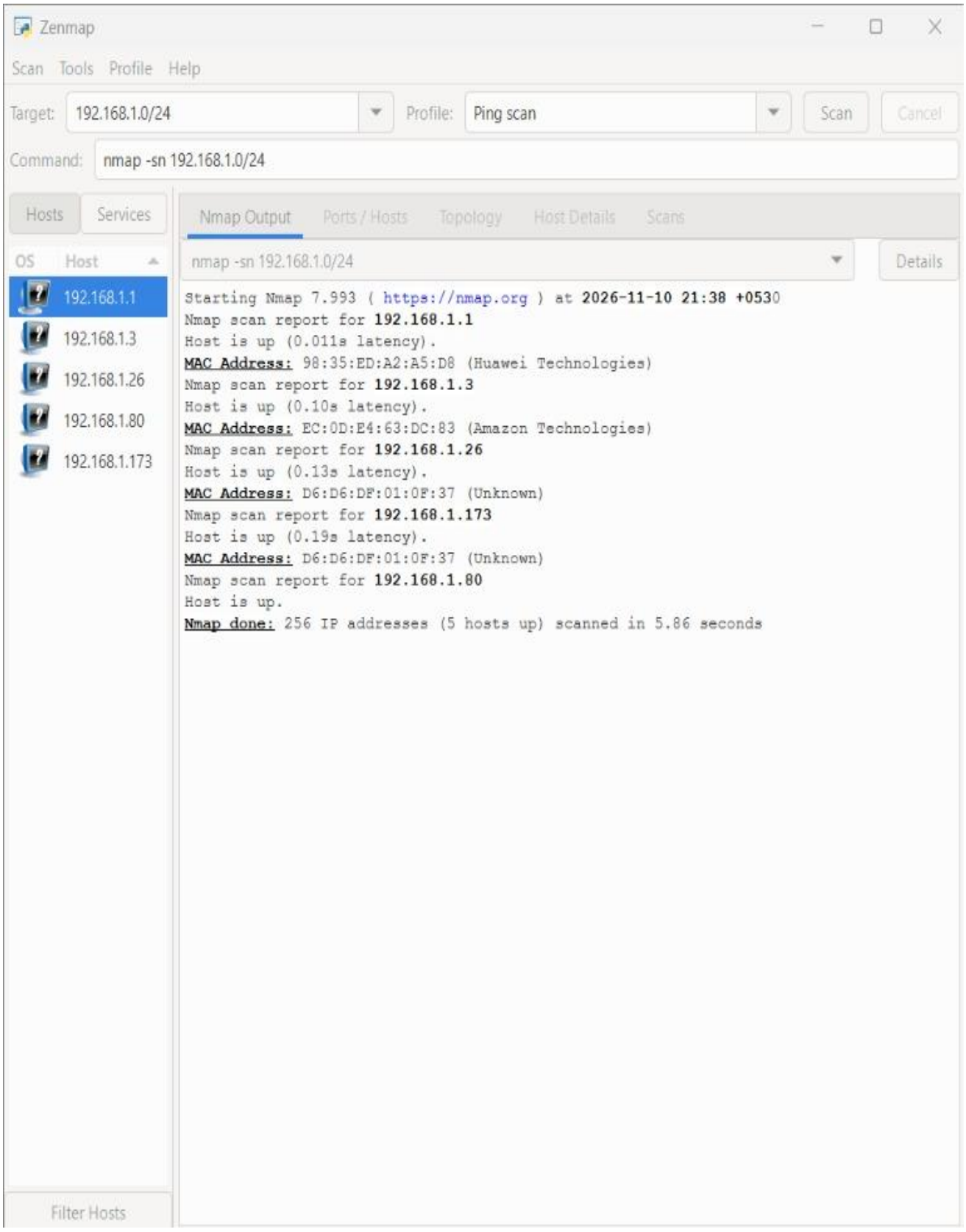
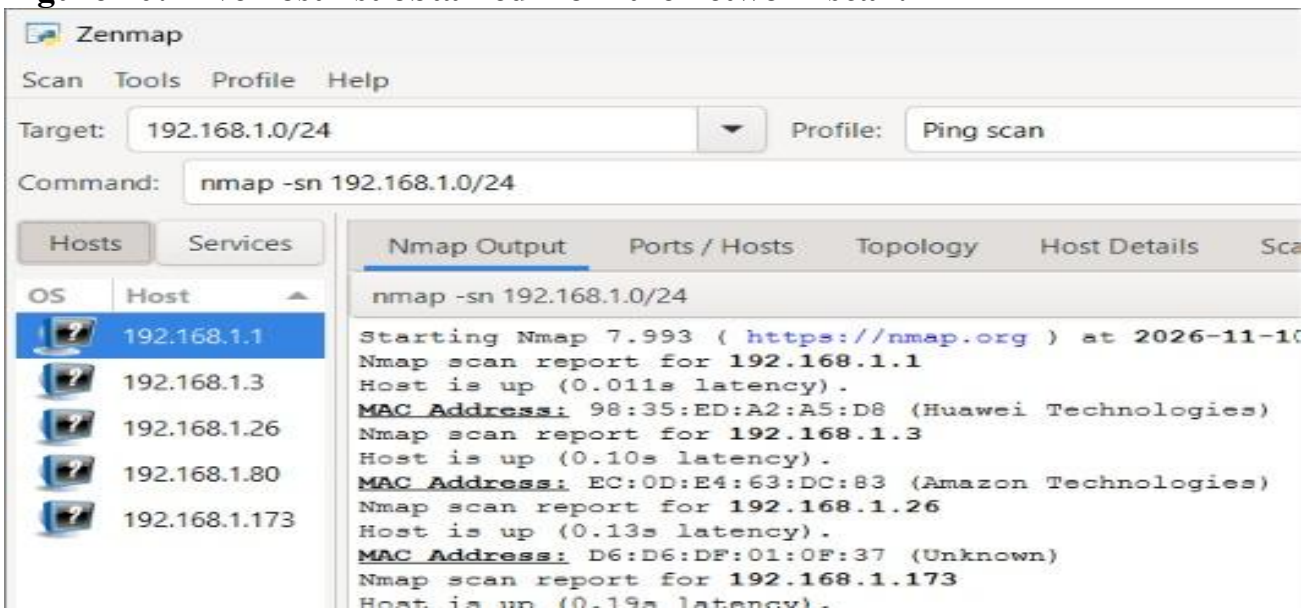

Zenmap
Scan Tools Profile Help
Target: 192.168.1.0/24
Profile: Ping scan
Scan
Cancel
Command: nmap -sn 192.168.1.0/24
Hosts
Services
OS Host
192.168.1.1
192.168.1.3
192.168.1.26
192.168.1.80
192.168.1.173
Filter Hosts
Nmap Output
Ports / Hosts
Topology
Host Details
Scans
nmap -sn 192.168.1.0/24
Details
Starting Nmap 7.993 (https://nmap.org) at 2026-11-10 21:38 +0530
Nmap scan report for 192.168.1.1
Host is up (0.011s latency).
MAC Address: 98:35:ED:A2:A5:D8 (Huawei Technologies)
Nmap scan report for 192.168.1.3
Host is up (0.10s latency).
MAC Address: EC:0D:E4:63:DC:83 (Amazon Technologies)
Nmap scan report for 192.168.1.26
Host is up (0.13s latency).
MAC Address: D6:D6:DF:01:0F:37 (Unknown)
Nmap scan report for 192.168.1.173
Host is up (0.19s latency).
MAC Address: D6:D6:DF:01:0F:37 (Unknown)
Nmap scan report for 192.168.1.80
Host is up.
Nmap done: 256 IP addresses (5 hosts up) scanned in 5.86 seconds

Figure 10: Live host list obtained from the network scan.



6.6 MAC Address Identification

MAC addresses were recorded from the available network scan information.

The local computer's MAC address was separately obtained using Windows Command Prompt.

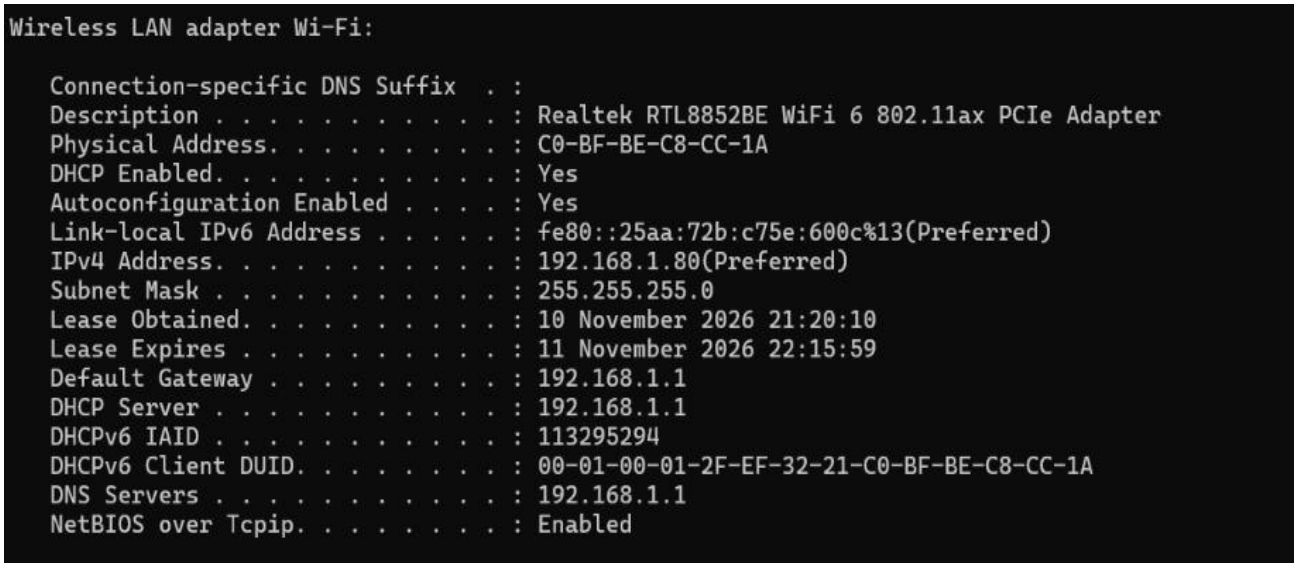
The local MAC address was:

`C0-BF-BE-C8-CC-1A`

MAC address information helps distinguish network interfaces and supports basic device identification within a local network.

It should be noted that MAC addresses can be randomized or changed, and vendor identification alone does not conclusively establish device ownership.

Figure 11: Local computer MAC address obtained using ipconfig /all.

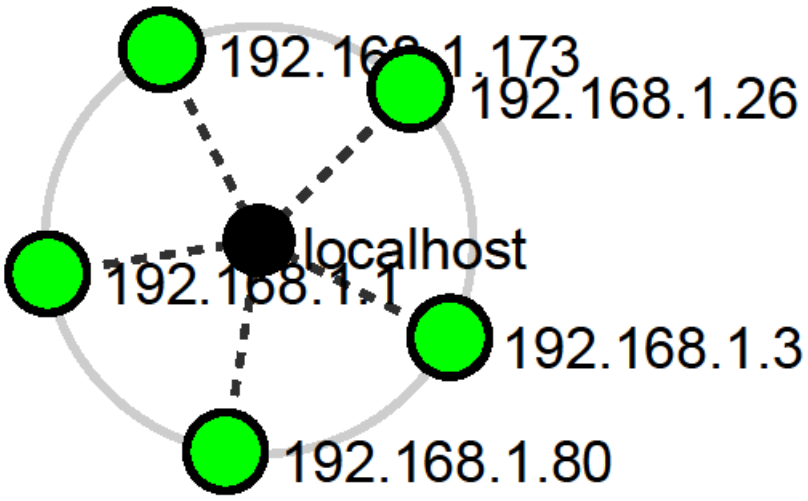


6.7 Network Topology

The Topology feature available in Zenmap was used to visualize the discovered hosts and their network relationships.

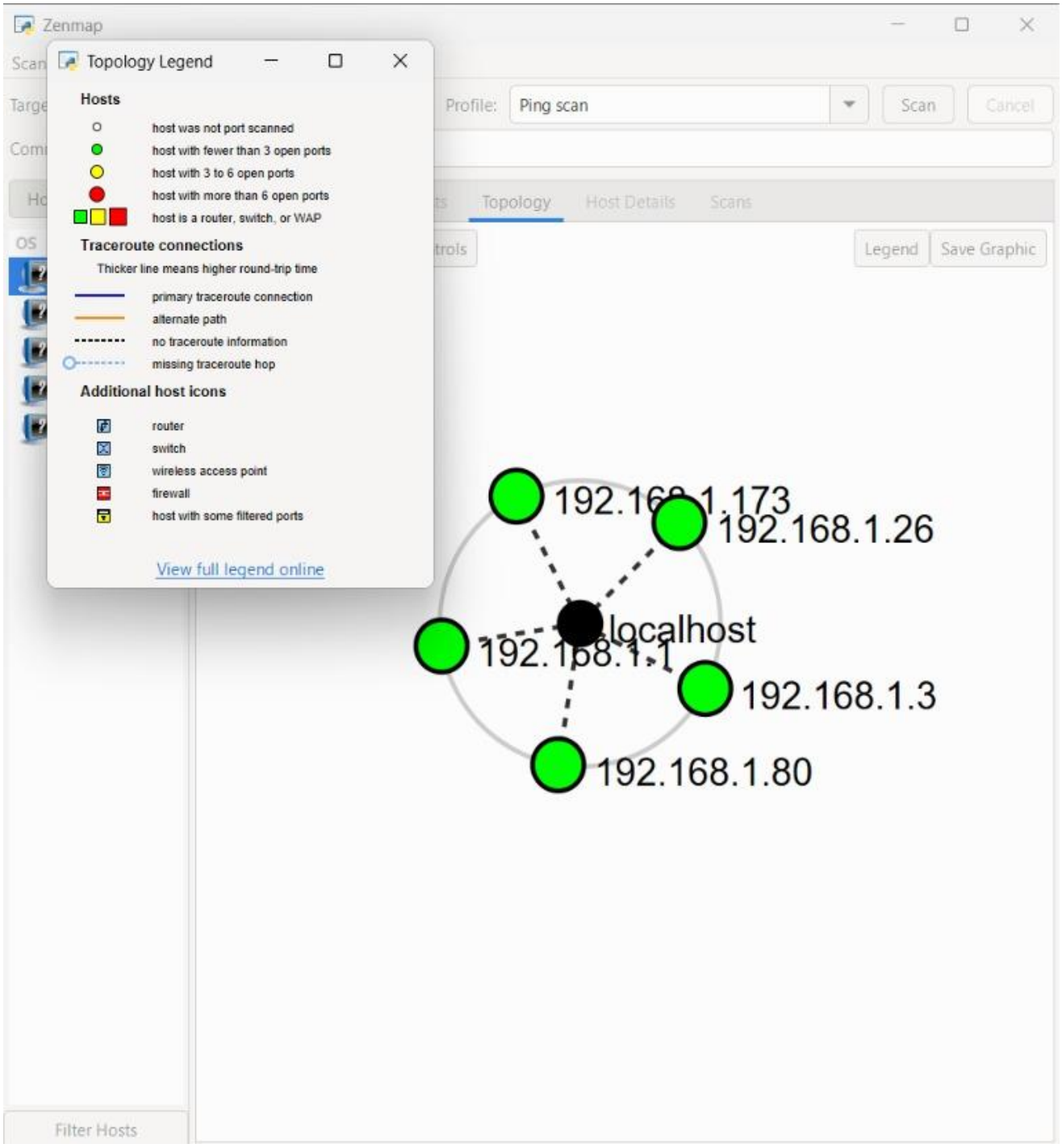
The resulting topology graph was successfully exported in PDF format.

Network_Topology.pdf



The topology visualization provided a graphical representation of the discovered network devices and helped demonstrate their relationships within the scanned local network.

Figure 12: Network topology generated using Zenmap.



The original exported PDF is included separately as supporting evidence.

7. RISK ANALYSIS

The observations collected during the practical exercises were reviewed to identify potential security considerations.

The following table presents preliminary observations and their possible security implications.

No.	Observation	Potential Security Implication	Suggested Risk
1	Public domain registration information	May assist infrastructure profiling	Low
2	Web technologies identified	May reveal software components requiring maintenance review	Medium
3	Public IP address identified	Provides infrastructure information	Low
4	HTTP headers expose application metadata	May assist technology fingerprinting	Low
5	WAF technology detected	Reveals part of the website's defensive architecture	Low
6	DNS records identified	May reveal mail and DNS infrastructure details	Medium
7	Multiple active LAN hosts	Requires device inventory and monitoring	Medium
8	Repeated MAC address in recorded results	Requires verification of device identification data	Informational

Note: These are preliminary educational assessments based on reconnaissance observations. No confirmed vulnerabilities were established by the tools used in these exercises.

8. SECURITY RECOMMENDATIONS

Based on the observations collected during the practical exercises, the following recommendations are suggested:

1. **Regular Software Updates:** Keep content management systems, plugins, frameworks and server software updated.
2. **Technology Information Review:** Periodically review publicly exposed technology information and remove unnecessary version disclosures where appropriate.
3. **HTTP Header Review:** Review HTTP response headers and implement appropriate security policies based on application requirements.
4. **DNS Configuration Review:** Regularly audit DNS records to ensure that only necessary and accurate records are publicly available.
5. **SPF Configuration:** Periodically review SPF records to ensure that authorized email-sending sources are correctly defined.
6. **WAF Maintenance:** Maintain and regularly review Web Application Firewall configurations and security rules.
7. **Network Device Inventory:** Maintain an updated inventory of devices connected to the local network.
8. **Network Monitoring:** Monitor network devices and investigate unexpected or unidentified hosts.
9. **Network Documentation:** Maintain updated network topology diagrams to improve infrastructure visibility and administration.

9. CONCLUSION

During Week 2 of the Cybersecurity and Ethical Hacking Internship Programme at Networkwalks, I successfully completed practical exercises covering Footprinting, Reconnaissance and Network Scanning.

In W2-PM1, six Kali Linux tools were used: WHOIS, WhatWeb, Nslookup, Curl, Wafw00f and DNSRecon. These tools helped collect publicly available domain registration information, identify web technologies, inspect HTTP response headers, detect a Web Application Firewall and enumerate DNS records.

In W2-PM5, Zenmap was used to perform network discovery on a local Wi-Fi network. The practical involved identifying network configuration details, discovering five active hosts, recording available IP and MAC address information, and generating a network topology.

These exercises provided practical experience in information gathering, web technology fingerprinting, DNS enumeration and network discovery.

Overall, the practical activities strengthened my understanding of fundamental penetration testing methodologies and demonstrated the importance of systematic information collection, technical documentation and security analysis during cybersecurity assessments.

10. EVIDENCE AND SCREENSHOTS

The following screenshots and files were collected as evidence of the completed practical exercises.

A. Kali Linux – W2-PM1

No.	Evidence	Filename
1	WHOIS Enumeration	01_WHOIS.png
2	WhatWeb Fingerprinting	02_WhatWeb.png
3	Nslookup DNS Resolution	03_Nslookup.png
4	Curl HTTP Header Analysis	04_Curl.png
5	Wafw00f Detection	05_Wafw00f.png
6	DNSRecon Enumeration	06_DNSRecon.png

B. Zenmap – W2-PM5

No.	Evidence	Filename
1	Local IP Configuration	01_IP_Configuration.png
2	Zenmap Scan Configuration	02_Scan_Configuration.png
3	Live Host Discovery	03_Live_Host_Discovery.png
4	Live Host List	04_Live_Host_List.png
5	Local MAC Address	05_Local_MAC_Address.png

6	Network Topology	06_Network_Topology.png
7	Exported Topology	Network_Topology.pdf

END OF REPORT

Author- Subhangi

NETWORKWALKS INTERNSHIP PROGRAMME

WEEK 02 – PENETRATION TESTING PRACTICALS